

ANSWERS

1 (a) (i)	$P = VI$	C1
	$60 = 12 \times I$	
	$I = 5.0 \text{ A}$	A1 [2]
	(ii) either $V = IR$ or $P = I^2 R$ or $P = V^2 / R$	C1
	either $12 = 5 \times R$ or $60 = 5^2 \times R$ or $60 = 12^2 / R$	M1
	$R = 2.4 \Omega$	A0 [2]
(b)	$R = \rho L / A$	C1
	$A = \pi \times (0.4 \times 10^{-3})^2 (= 5.03 \times 10^{-7})$	C1
	$L = (2.4 \times 5.03 \times 10^{-7}) / (1.0 \times 10^{-6})$	
	$= 1.2 \text{ m}$	A1 [3]
(c)	resistance is halved	M1
	either current is doubled or power $\propto 1/R$	M1
	power is doubled	A1 [3]
2 (a)	ohm = volt / ampere	B1 [1]
(b)	$\rho = RA / l$ or unit is $\Omega \text{ m}$	C1
	units: $\text{VA}^{-1} \text{m}^2 \text{m}^{-1} = \text{NmC}^{-1} \text{A}^{-1} \text{m}^2 \text{m}^{-1}$	C1
	$= \text{kgm}^2 \text{s}^{-2} \text{A}^{-1} \text{s}^{-1} \text{A}^{-1} \text{m}^2 \text{m}^{-1}$	
	$= \text{kgm}^3 \text{s}^{-3} \text{A}^{-2}$	A1 [3]
(c) (i)	$\rho = [3.4 \times 1.3 \times 10^{-7}] / 0.9$	C1
	$= 4.9 \times 10^{-7} (\Omega \text{ m})$	A1 [2]
	(ii) max = 2.0 V	A1
	min = $2 \times (3.4 / 1503.4) = 4.5 \times 10^{-3} \text{ V}$	A1 [2]
	(iii) $P = V^2 / R$ or $P = VI$ and $V = IR$	C1
	$= (2)^2 / 3.4$	
	$= 1.18 \text{ (allow 1.2) W}$	A1 [2]
(d) (i)	power in Q is zero when $R = 0$	B1 [1]
	(ii) power in Q = 0 / tends to zero as $R = \text{infinity}$	B1 [1]
3 (a) (i)	$R = V^2 / P$ or $P = IV$ and $V = IR$	C1
	$= (220)^2 / 2500$	
	$= 19.4 \Omega \text{ (allow 2 s.f.)}$	A1 [2]
	(ii) $R = \rho l / A$	C1
	$l = [19.4 \times 2.0 \times 10^{-7}] / 1.1 \times 10^{-6}$	C1
	$= 3.53 \text{ m (allow 2 s.f.)}$	A1 [3]
(b) (i)	$P = 625, 620 \text{ or } 630 \text{ W}$	A1 [1]
	(ii) R needs to be reduced	C1
	Either length $\frac{1}{4}$ of original length	
	or area $4 \times$ greater	
	or diameter $2 \times$ greater	A1 [2]

4 (a)	(i) $I_1 = I_2 + I_3$	B1	[1]
	(ii) $I = V / R$ $R = [1/6 + 1/10]^{-1}$ [total $R = 3.75 \Omega$] $I_1 = 12 / 3.75 = 3.2 \text{ A}$	C1	
	or $I_2 = 12 / 10$ (= 1.2 A) or $I_3 = 12 / 6$ (= 2.0 A) or $I_1 = 1.2 + 2.0 = 3.2 \text{ A}$	C1	
		A1	[3]
	(iii) power = VI or I^2R or V^2 / R	C1	
	$x = \frac{\text{power in wire}}{\text{power in series resistors}} = \frac{I_2^2 R_w}{I_3^2 R_s}$ or $\frac{VI_2}{VI_3}$ or $\frac{V^2 / R_w}{V^2 / R_s}$	C1	
	$x = 12 \times 1.2 / 12 \times 2.0 = 0.6(0)$ allow 3 / 5 or 3:5	A1	[3]
	(b) p.d. BC: $12 - 12 \times 0.4 = 7.2 \text{ (V)}$ / p.d. AC = 4.8 (V) p.d. BD: $12 - 12 \times 4 / 6 = 4.0 \text{ (V)}$ / p.d. AD = 8.0 (V) p.d. = 3.2 V	C1	
		C1	
		A1	[3]
5	(a) resistance = potential difference / current	B1	[1]
	(b) (i) metal wire in series with power supply and ammeter voltmeter in parallel with metal wire	B1	
	rheostat in series with power supply or potential divider arrangement or variable power supply	B1	
	(ii) 1. intercept on graph	B1	[1]
	2. scatter of readings about the best fit line	B1	[1]
	(iii) correction for zero error explained use of V and corrected I values from graph resistance = $V / I = 22.(2) \Omega$ [e.g. $4.0 / 0.18$]	B1	
		C1	
		A1	[3]
	(c) $R = 6.8 / 0.64 = 10.625$	C1	
	%R = %V + %I = $(0.1 / 6.8) \times 100 + (0.01 / 0.64) \times 100$ = 1.47% + 1.56%	C1	
	$\Delta R = 0.0303 \times 10.625 = 0.32 \Omega$		
	$R = 10.6 \pm 0.3 \Omega$	A1	[3]

6	(a) p.d. = work (done) / charge OR energy transferred from (electrical to other forms) / (unit) charge	B1	[1]
(b)	(i) $R = \rho l / A$ $\rho = 18 \times 10^{-9}$ $R = (18 \times 10^{-9} \times 75) / 2.5 \times 10^{-6} = 0.54 \Omega$	C1	
		C1	
		A1	[3]
	(ii) $V = IR$ $R = 38 + (2 \times 0.54)$ $I = 240 / 39.08 = 6.1 \text{ (6.14) A}$	C1	
		C1	
		A1	[3]
	(iii) $P = I^2 R$ or $P = VI$ and $V = IR$ or $P = V^2 / R$ and $V = IR$ $= (6.14)^2 \times 2 \times 0.54$ $= 41 \text{ (40.7) W}$	C1	
		C1	
		A1	[3]
(c)	area of wire is less (1/5) hence resistance greater ($\times 5$) OR R is $\propto 1/A$ therefore R is greater p.d. across wires greater so power loss in cables increases	M1	
		A1	[2]
7	(a) e.m.f. = total energy available (per unit charge) some (of the available energy) is used/lost/wasted/given out in the internal resistance of the battery (hence p.d. available less than e.m.f.)	B1	
		B1	[2]
(b)	(i) $V = IR$ $I = 6.9 / 5.0 = 1.4 \text{ (1.38) A}$	C1	
		A1	[2]
	(ii) $r = \text{lost volts} / \text{current}$ $= (9 - 6.9) / 1.38 = 1.5(2) \Omega$	C1	
		A1	[2]
(c)	(i) $P = EI$ (not $P = VI$ if only this line given or 9 V not used in second line) $= 9 \times 1.38 = 12 \text{ (12.4) W}$	C1	
		A1	[2]
	(ii) efficiency = output power / total power $= VI / EI = 6.9 / 9$ or $(9.52) / (12.4) = 0.767 / 76.7\%$	C1	
		A1	[2]
8	(a) e.m.f.: energy converted from chemical/other forms to electrical per unit charge p.d.: energy converted from electrical to other forms per unit charge	B1	
		B1	[2]
(b)	(i) the p.d. across the lamp is <u>less than</u> 12 V or there are lost volts/power/energy in the battery/internal resistance	B1	[1]
	(ii) $R = V^2 / P$ (or $V = RI$ <u>and</u> $P = VI$) $= 144 / 48$ $= 3.0 \Omega$	C1	
		A1	[2]
	(iii) $I = E / (R_T + r)$ $= 12 / 2.0$ $= 6.0 \text{ A}$	C1	
		A1	[2]
	(iv) power of each lamp = $I^2 R$ $= (3.0)^2 \times 3.0$ $= 27 \text{ W}$	C1	
		A1	[2]
(c)	less resistance (in circuit)/more current more lost volts/less p.d. across battery	M1	
		A1	[2]

9	(a) lost volts/energy used within the cell/internal resistance when cell supplies a current	B1 B1 [2]
	(b) (i) $E = I(R + r)$ $4.5 = 0.65(6.0 + r)$ $r = 0.92\Omega$	C1 A1 [2]
	(ii) $I = 0.65$ (A) and $V = IR$ $V = 0.65 \times 6 = 3.9\text{V}$	C1 A1 [2]
	(iii) $P = V^2/R$ or $P = I^2R$ and $P = IV$ $= (3.9)^2/6 = 2.5\text{W}$	C1 A1 [2]
	(iv) efficiency = power out/power in $= I^2R/I^2(R + r) = R/(R + r) = 6.0/(6.0 + 0.92) = 0.87$	C1 A1 [2]
	(c) (circuit) resistance decreases current increases more heating effect	B1 M1 A1 [3]
10	(a) <u>random</u> error (in the measurements) of the length OR resistance	B1 [1]
	(b) gradient = $(3.6 - 1.9) / (0.8 - 0.4)$ $= 4.25$	C1 A1 [2]
	(c) $R = \rho l / A$ $\rho = \text{gradient} \times \text{area} = 4.25 \times 0.12 \times 10^{-6}$ $= 5.1(0) \times 10^{-7} \Omega\text{m}$	C1 C1 A1 [3]
	(d) resistance decreasing with increasing area correct shape with curve being asymptote to both axes	B1 B1 [2]
11	(a) very high/infinite resistance for negative voltages up to about 0.4V resistance decreases from 0.4V	B1 B1 [2]
	(b) initial straight line from (0,0) into curve with decreasing gradient but not to horizontal repeated in negative quadrant	M1 A1 [2]
	(c) (i) $R = 12^2/36 = 4.0\Omega$ or $I = P/V = 36/12 = 3.0\text{A}$ and $R = 12/3.0 = 4.0\Omega$	A1 (A1) [1]
	(ii) lost volts = $0.5 \times 2.8 = 1.4$ (V) or $E = 12 = 2.8 \times (R + r)$	C1
	$R = V/I = (12 - 1.4)/2.8$ or $(R + r) = 4.29\Omega$	C1
	$= 3.8$ (3.79) Ω or $R = 3.8\Omega$	A1 [3]
	(d) resistance of the lamp increases with increase of V or I	B1 [1]

12 (a)	(i) movement/flow of charge carriers	B1	
	(ii) $\frac{\text{work (done) or energy (transformed)(from electrical to other forms)}}{\text{charge}}$	B1	
(b)	(i) p.d. across one lamp = 2.5 V resistance = $[(8.7 - 7.5)/0.3]/2 = 2.0 (\Omega)$	C1	
	(ii) straight line through the origin with gradient of 0.5	M1	
		A1	
	(iii) $P = I^2 R$ or $P = VI$ and $V = IR$ or $P = V^2 / R$ and $V = IR$ = $0.30^2 \times 2.0$ = 0.60×0.30 = $0.60^2 / 2.0$ = 0.18 (W)	C1	
		A1	
(iv)	1 $R = \rho l / A$ $l = (2.0 \times 0.40 \times 10^{-6}) / 1.7 \times 10^{-8}$ = 47 (m)	C1	
		A1	
	2 $I = Anvq$ $v = 0.30 / (0.40 \times 10^{-6} \times 8.5 \times 10^{28} \times 1.6 \times 10^{-19})$ = $5.5 \times 10^{-5} (\text{m s}^{-1})$	C1	
		A1	
13 (a)	(coulomb is) ampere second	B1	[1]
	(b) (total) charge or $Q = nAle$	M1	
	$I = Q/t$ and $l/t = v$	M1	
	$I = nAle/t = nAve$ therefore $v = I/nAe$	A1	[3]
(c)	(i) ratio = $(I/nA_Y e)/(I/nA_Z e)$ = A_Z/A_Y or $4A/A$ or $\pi d^2/(\pi d^2/4)$ = 4	C1	
		C1	
		A1	[3]
	(ii) $R = \rho l/A$ or $R = 4\rho l/\pi d^2$	B1	
	$R_Y = \rho l/A$ and $R_Z = \rho(2l)/4A$ so $R_Y/R_Z = 2$ or $R_Y = 4\rho l/\pi d^2$ and $R_Z = 4\rho(2l)/\pi 4d^2$ or $2\rho l/\pi d^2$ so $R_Y/R_Z = 2$	A1	[2]
(iii)	$V = 12R_Y/(R_Y + R_Z)$ or $I = 12/(R_Y + R_Z)$ and $V = IR_Y$ $V = 12 \times 2/3$ = 8(0) V	C1	
		A1	[2]
(iv)	ratio = $I^2 R_Y/I^2 R_Z$ or $(V_Y^2/R_Y)/(V_Z^2/R_Z)$ or $(V_Y I)/(V_Z I)$ = 2	A1	[1]

14 (a)	charge exists only in discrete amounts	B1	[1]
(b) (i)	$E = I(R + r)$ or $V = IR$	C1	
	(total resistance =) $2.7 + 0.30 + 0.25 (= 3.25 \Omega)$	M1	
	$I = 9.0 / (2.7 + 0.30 + 0.25)$ or $9.0 / 3.25 = 2.8 \text{ A}$	A1	[3]
(ii)	$V = IR_{\text{ext}}$	C1	
	$= 2.77 \times 3.0$ or 2.8×3.0		
	or		
	$V = E - Ir$	(C1)	
	$= 9.0 - 2.77 \times 0.25$ or $9.0 - 2.8 \times 0.25$		
	$V = 8.3 (8.31) \text{ V}$ or 8.4 V	A1	[2]
(c) (i)	$I = nevA$		
	$v = 2.77 / (8.5 \times 10^{29} \times 1.6 \times 10^{-19} \times 2.5 \times 10^{-6})$	M1	
	$= 8.1 (8.147) \times 10^{-6} \text{ ms}^{-1}$ or $8.2 \times 10^{-6} \text{ ms}^{-1}$	A1	[2]
(ii)	A reduces by a factor 4 (1/4 less) or resistance of Z goes up by 4×	M1	
	current goes down but by <u>less than</u> a factor of 4 (as total resistance does not go up by a factor of 4) so drift speed goes up	A1	[2]
15 (a)	ohm is volt per ampere or volt/ampere	B1	[1]
(b) (i)	$R = \rho l / A$	B1	
	$R_P = 4\rho(2l) / \pi d^2$ or $8\rho l / \pi d^2$ or $R_Q = \rho l / \pi d^2$		
	or		
	ratio idea e.g. length is halved hence R halved and diameter is halved hence R is 1/4	C1	
	$R_Q (= 4\rho l / \pi 4d^2) = \rho l / \pi d^2$		
	$= R_P / 8$		
	$(= 12/8) = 1.5 \Omega$	A1	[3]
(ii)	power = $I^2 R$ or V^2 / R or VI	C1	
	$= (1.25)^2 \times 12 + (10)^2 \times 1.5$ or $(15)^2 / 12 + (15)^2 / 1.5$ or 15×11.25	C1	
	$= (18.75 + 150) = 170 (168.75) \text{ W}$	A1	[3]
(iii)	$I_P = (15 / 12) = 1.25 \text{ (A)}$ and $I_Q = (15 / 1.5) = 10 \text{ (A)}$	C1	
	$v_P / v_Q = I_P n A_Q e / I_Q n A_P e$ or $(1.25 \times \pi d^2) / (10 \times \pi d^2 / 4)$	C1	
	$= 0.5$	A1	[3]

16 (i) $\rho = (RA/l)$	C1
$= (0.03 \times 1.5 \times 10^{-6}) / 2.5 (= 1.8 \times 10^{-8})$	C1
$= 18 \text{ n}\Omega \text{ m}$	A1
17 (a) (the ohm is) volt / ampere	B1
(b) (i) $R = \rho L / A$	C1
ratio $= [\rho L / (\pi d^2 / 4)] / [0.028\rho \times 7.0L / \{\pi(14d)^2 / 4\}] = 1000$	A1
or ratio $= 14^2 / (0.028 \times 7) = 1000$	
(ii) same current (in connecting and filament wires) and the lamp/filament (wire) has greater resistance	B1
(iii) $P = V/R$ or $P = VI$ or $P = I R$	C1
(for filament wire) $R = 12^2 / 6.0$ or $R = 6.0 / 0.50^2$ or $R = 12 / 0.50$	C1
(for filament wire) $R = 24 \Omega$	A1
(for connecting wire) $R = 24 / 1000$	
$= 2.4 \times 10^{-2} \Omega$	
(iv) resistance of connecting wire increases	B1
current in circuit/lamp/filament (wire) decreases	M1
or potential difference across lamp/filament (wire) decreases	
(so) resistance of lamp/filament (wire) decreases	A1
18 (a) flow of charge carriers	B1
(b) (i) $nALe$	B1
(ii) t is time taken for electrons to move length L)	B1
$I = Q/t$	
$I = nALe/t$ or $I = nALe/(L/v)$ or $I = nAvte/t$ and $I = nAve$	B1
(c) (i) ratio = area at X/area at Y	C1
$= [\pi d^2 / 4] / [\pi(0.69d)^2 / 4]$ or $d^2 / (0.69d)^2$ or $1 / 0.69^2$	
$= 2.1$	A1
(ii) 1. $R = \rho L / A$ or $R/L \propto 1/A$	C1
resistance per unit length $= 1.7 \times 10^{-2} \times (\text{area at X/area at Y})$	A1
$= 1.7 \times 10^{-2} \times 2.1$	
$= 3.6 \times 10^{-2} \Omega \text{ m}^{-1}$	
2. $P = I^2 R$ or $P = V^2 / R$	C1
$R = 3.6 \times 10^{-2} \times 3.0 \times 10^{-3} (= 1.08 \times 10^{-4} \Omega)$	A1
$P = 0.50^2 \times 1.08 \times 10^{-4}$ or $P = (5.4 \times 10^{-5})^2 / 1.08 \times 10^{-4}$	
$= 2.7 \times 10^{-5} \text{ W}$	
(iii) (cross-sectional area decreases so) resistance increases	M1
$(P = I^2 R, \text{ so})$ power increases	A1

19 (a) (i)	$P = VI$	C1	(ii)	$Q = 0.25 \times 3.0 \times 3600 (= 2700)$	C1	
	$I = 30 / 120$	A1		number = $(0.25 \times 3.0 \times 3600) / 1.60 \times 10^{-19}$	A1	
	$= 0.25 \text{ A}$			$= 1.7 \times 10^{22}$		
(b)	$R = V / I$	or	$R = P^2 / I$	or	$R = V^2 / P$	C1
	$= 120 / 0.25$	or	$= 30 / 0.25^2$	or	$= 120^2 / 30 = 480 \Omega$	A1
(c)	$R = \rho l / A$					C1
	$A = (6.1 \times 10^{-7} \times 580 \times 10^{-3}) / 480 (= 7.37 \times 10^{-10})$					C1
	$d = [(4 \times 7.37 \times 10^{-10}) / \pi]^{1/2}$					A1
	$= 3.1 \times 10^{-5} \text{ m}$					
(d)	temperature decreases and so resistance decreases					B1

20 (a)	joule /coulomb	B1
(b) (i)	lamps have same p.d./lamps have p.d. of 2.7 V	B1
	current = $0.15 + 0.090$	A1
	$= 0.24 \text{ A}$	
(ii)	$R = (4.5 - 2.7) / 0.24$	C1
	or	
	$R_P = 18 (\Omega)$ and $R_Q = 30 (\Omega)$	
	$1/R_T = 1/18 + 1/30$ and so $R_T = 11.25$	
	$4.5 = 0.24 \times (R + 11.25)$	
	$R = 7.5 \Omega$	A1
(iii)	$R = \rho l / A$	C1
	$R_P / R_Q = [(2.7 / 0.15) / (2.7 / 0.09)] (= 0.60)$	C1
	ratio = 0.60×2^2	A1
	$= 2.4$	
(iv)	less p.d. across resistor/greater p.d. across <u>P</u>	B1
	greater current through <u>P</u> and so resistance (of P) increases	B1

21 (a)	$R = \rho L / A$	C1
	$3.0 = \rho / [\pi \times (0.48 \times 10^{-3} / 2)^2]$	C1
	$\rho = 5.4 \times 10^{-7} \Omega \text{m}$	A1
(b) (i)	p.d. = $5.0 - (2.0 \times 1.6)$ = 1.8V	A1
(ii) 1.	current in resistor = $1.8 / 4.5$ (= 0.40 A)	C1
	current in wire = $1.6 - 0.40$ (= 1.2 A)	C1
	$R_X = 1.8 / 1.2$ = 1.5 Ω	A1
	or $R_T = 1.8 / 1.6$ or $(5.0 / 1.6) - 2.0$ (= 1.125 Ω)	(C1)
	$(1 / 1.125) = (1 / 4.5) + (1 / R_X)$	(C1)
	$R_X = 1.5 \Omega$	(A1)
2.	length = $1.5 / 3.0$ or $1.5 \times 1.8 \times 10^{-7} / (5.4 \times 10^{-7})$ = 0.50 m	A1

22 (a)	A: (cross-sectional) area (of wire)	B1
	n : number of free electrons per unit volume or number density of free electrons	B1
(b)	line drawn between (X, v_x) and (Y, $4v_x$)	M1
	line has increasing gradient	A1

23 (a)	(coulomb is an) ampere second	B1
(b)	$8.0 \times 10^{-19} \text{C}$ and $1.6 \times 10^{-19} \text{C}$ both underlined (and no others underlined)	B1
(c)	line drawn between (S, $1.00v_s$) and (T, $0.25v_s$)	M1
	line with decreasing magnitude of gradient	A1

24 (a)	volt / ampere	B1
(b) (i)	1. $I = 1.8 + 0.60$ $= 2.4 \text{ A}$ 2. $(8.0 \times 0.60) = 1.8 \times (2.0 + R_Z)$ $R_Z = 0.67 \Omega$ 3. $E - (2.4 \times 1.5) = (0.60 \times 8.0)$ or $E - (2.4 \times 1.5) = 1.8 \times (2.0 + 0.67)$ or $E = 2.4 \times [1.5 + (8.0 \times 2.67) / (8.0 + 2.67)]$ $E = 8.4 \text{ V}$	A1 C1 A1 C1 A1
(ii)	1. $R = \rho L / A$ or $R \propto 1 / A$ ratio $= R_Y / R_X = 2.0 / 8.0$ $= 0.25$ 2. $I \propto Av$ or $I_X / I_Y = A_X v_X / A_Y v_Y$ ratio $= (0.60 / 1.8) \times (1 / 0.25)$ $= 1.3$	C1 A1 C1 A1

25 (a)	A : cross-sectional area	B1
	n : number density of <u>free</u> electrons	B1
(b)	units of I : A and units of A : m^2 and units of v : m s^{-1}	B1
	units of e : $\text{A} / (\text{m}^2 \text{m}^{-3} \text{m s}^{-1}) = \text{A s}$	A1
(c)	ratio $= A_Q / A_P$	C1
	$= [\pi r^2] / [\pi (2r^2)]$	A1
	$= 0.25$	

26.

6(a)	$I = I_1 + I_2 + I_3$	B1
	$(V/R) = (V/R_1) + (V/R_2) + (V/R_3)$ or $(I/V) = (I_1/V) + (I_2/V) + (I_3/V)$	B1
	and $1/R = 1/R_1 + 1/R_2 + 1/R_3$	
6(b)(i)	current = 0.49 + 0.45 = 0.94 A	A1
6(b)(ii)	$8.0 = (0.94 \times r) + (0.45 \times 16)$	C1
	$r = 0.85 \Omega$	A1
6(c)	$I = Anvq$	C1
	$v = (0.45/0.49) \times 2.1 \times 10^{-4}$	
	$= 1.9 \times 10^{-4} \text{ m s}^{-1}$	A1
6(d)	total/combined resistance decreases	B1
	(current in battery increases so terminal) potential difference decreases	B1

27.

5(a)	volt / ampere	B1
5(b)	$R = \rho L / A$	B1
	$(A = V / L)$	B1
	(so) $R = \rho L^2 / V$ (with ρ and V constant so $R \propto L^2$)	
5(c)(i)	$E = 2.4 \text{ V}$	A1
5(c)(ii)	$P = VI$ or $I^2 R$ or V^2 / R	C1
	$= 1.3 \times 5.0$ or $5.0^2 \times 0.26$ or $1.3^2 / 0.26$	C1
	$= 6.5 \text{ W}$	A1
5(c)(iii)	(-) internal resistance or (-) r	B1

28.

6(a)	$Q = It$	C1
	$= 0.80 \times 7.5 \times 60$	A1
	$= 360 \text{ C}$	
6(b)	$P = EI$ or $P = VI$ or $P = I^2 R$ or $P = V^2 / R$	C1
	$0.80^2 \times 0.40 (= 0.256 \text{ W})$	C1
	or $0.48 \times 0.80 (= 0.384 \text{ W})$	
	efficiency = $(0.256 / 0.384) \times 100$ $= 67\%$	A1
6(c)(i)	$n = 3.2 \times 10^{22} / (1.3 \times 10^{-7} \times 3.0)$ $= 8.2 \times 10^{28} \text{ m}^{-3}$	A1
6(c)(ii)	$I = Anvq$	C1
	$v = 0.80 / (1.3 \times 10^{-7} \times 8.2 \times 10^{28} \times 1.60 \times 10^{-19})$	
	$= 4.7 \times 10^{-4} \text{ m s}^{-1}$	A1

6(d)	(wire Y has) larger resistance / resistance increases	M1
	(wire Y has) smaller current / current decreases	M1
	(average drift) speed is less (in wire Y)	A1

29.

6(a)(i)	$P = VI$	C1
	$I = 36 / 8.0$ $= 4.5 \text{ A}$	A1
6(a)(ii)	charge $= 4.5 \times 50$ $= 225$	C1
	number $= 225 / 1.6 \times 10^{-19}$ $= 1.4 \times 10^{21}$	A1
6(a)(iii)	$R = V^2 / P$ or $R = V / I$ or $R = P / I^2$ $= 8.0^2 / 36$ or $= 8.0 / 4.5$ or $= 36 / 4.5^2$	C1
	$= 1.8 \Omega$	A1
6(b)	$R = \rho L / A$	C1
	$L = (1.8 \times 0.25 \times 10^{-6}) / 1.4 \times 10^{-6}$ $= 0.32 \text{ m}$	A1
6(c)	(larger cross-sectional area, same length, same resistivity and so) less resistance	M1
	(same p.d. and more current so) more power (dissipated)	A1
6(d)	current (in wire) is the same	M1
	(same p.d. across wire so) power stays the same	A1

30.

6(a)	$V = V_1 + V_2 + V_3$	B1
	$IR = IR_1 + IR_2 + IR_3$ or $(V/I) = (V_1/I) + (V_2/I) + (V_3/I)$	B1
	and $R = R_1 + R_2 + R_3$	
6(b)(i)	$V / 8.0 = 6.0 \times 10^3 / (4.0 \times 10^3 + 6.0 \times 10^3)$	C1
	or $I = 8.0 / (4.0 \times 10^3 + 6.0 \times 10^3) = 8.0 \times 10^{-4}$	
	$V = 8.0 \times 10^{-4} \times 6.0 \times 10^3$	
	$V = 4.8 \text{ V}$	A1
6(b)(ii)	total resistance in parallel $= 3.0 \times 10^3 (\Omega)$ or $3.0 (\text{k}\Omega)$	C1
	current $= 8.0 / (3.0 \times 10^3 + 6.0 \times 10^3)$ $= 8.9 \times 10^{-4} \text{ A}$	A1
6(c)	<u>thermistor</u> resistance increases	B1
	(thermistor resistance increases so total resistance increases so) current decreases (in battery)	M1
	$(P = EI \text{ and } E \text{ constant so})$ power decreases	A1

31.

6(a)(i)	line passes through (0,0) and is in first and third quadrants only	M1																														
	gradient of line becoming less steep in both quadrants and roughly symmetrical	A1																														
6(a)(ii)	(as I increases) the temperature (of the filament wire/lamp) increases	B1																														
	(as I / temperature / V increases) the resistance (of wire/lamp) increases	B1																														
	(as I / temperature / V increases the graph curves because) ratio V / I increases or ratio I / V decreases	B1																														
6(b)	$R = \rho L / A$	C1																														
	$= (5.6 \times 10^{-8} \times 5.8) / 3.4 \times 10^{-8}$ $= 9.6 \Omega$	A1																														
6(c)	<table><tr><th colspan="2">position of switches</th><th colspan="3">ammeter readings</th></tr><tr><th>S_1</th><th>S_2</th><th>X / A</th><th>Y / A</th><th>Z / A</th></tr><tr><td>open</td><td>open</td><td>0</td><td>0</td><td>0</td></tr><tr><td>open</td><td>closed</td><td>1.0</td><td>0</td><td>1.0</td></tr><tr><td>closed</td><td>open</td><td>2.0</td><td>2.0</td><td>0</td></tr><tr><td>closed</td><td>closed</td><td>3.0</td><td>2.0</td><td>1.0</td></tr></table> second row: both values correct (B1) third row: all three values correct (B1) fourth row: $X = Y + Z$ (any values) (B1) all three values correct (B1)	position of switches		ammeter readings			S_1	S_2	X / A	Y / A	Z / A	open	open	0	0	0	open	closed	1.0	0	1.0	closed	open	2.0	2.0	0	closed	closed	3.0	2.0	1.0	B4
position of switches		ammeter readings																														
S_1	S_2	X / A	Y / A	Z / A																												
open	open	0	0	0																												
open	closed	1.0	0	1.0																												
closed	open	2.0	2.0	0																												
closed	closed	3.0	2.0	1.0																												

32.

6(a)	energy transferred per (unit) charge (from electrical to other forms)	B1
6(b)	same/equal current (in X and Y)	B1
	$P = I^2 R$ (and $R_X > R_Y$) or $P = VI$ and $V_X > V_Y$	M1
	(so) X (dissipates more power)	A1
6(c)(i)	$I = Q / t$	C1
	$= 650 / 540$	A1
	$= 1.2 \text{ A}$	
6(c)(ii)	$V = W / Q$ or W / It	C1
	$= 4800 / 650$ or $4800 / (1.2 \times 540)$	A1
	$= 7.4 \text{ V}$	
	or	
	$V = P / I$ and $P = W / t$	(C1)
	$= 8.9 / 1.2$ $= 7.4 \text{ V}$	(A1)
6(c)(iii)	$I = 4.5 + 1.2 (= 5.7 \text{ A})$	C1
	$9.0 = 7.4 + 5.7r$ or $9.0 = 5.7 (1.3 + r)$	A1
	$r = 0.28 \Omega$	

33.

6(a)	energy (transferred from electrical to other forms) per unit charge	B1
6(b)(i)	(resistance is) infinite / very high	B1
6(b)(ii)	(resistance) decreases (as V increases)	B1
6(c)(i)	current = $2.7 - 1.5$ = 1.2 A	A1
6(c)(ii)	$12 = (1.5 \times 5.0) + (1.5 \times R)$ or $R = (12/1.5) - 5.0$	C1
	$R = 3.0 \Omega$	A1
6(c)(iii)	$V_{(XZ)} = (1.6 / 2.0) \times 12 (= 9.6 \text{ V})$	C1
	$V_{(XW)} = 1.5 \times 5.0 (= 7.5 \text{ V})$	C1
	potential difference = $9.6 - 7.5$ = 2.1 V	A1
	or	
	$V_{(ZY)} = (0.4 / 2.0) \times 12 (= 2.4 \text{ V})$	(C1)
	$V_{(WY)} = 1.5 \times 3.0 (= 4.5 \text{ V})$	(C1)
	potential difference = $4.5 - 2.4$ = 2.1 V	(A1)
6(c)(iv)	current in (fixed / variable) resistor decreases	B1
	current in (resistance) wire is unchanged	B1
	(so) current in battery decreases, (same e.m.f. so) power decreases	B1

34.

6(a)	(number density =) $2.3 \times 10^{23} / (1.5 \times 10^{-6} \times 1.8) = 8.5 \times 10^{28} \text{ (m}^{-3}\text{)}$	A1
6(b)	$I = Q / t$ or $I = 172 / 2.5 \times 60$ or $I = 1.1(5)$	C1
	$I = nAvq$	C1
	$172 / (2.5 \times 60) = 1.5 \times 10^{-6} \times 8.5 \times 10^{28} \times v \times 1.6 \times 10^{-19}$	
	$v = 5.6 \times 10^{-5} \text{ m s}^{-1}$	A1

35.

6(a)	temperature decreases (so) resistance decreases	B1
6(b)(i)	current = V / R	A1
6(b)(ii)	$I = Anvq$ $n = N / V$ or $n = N / AL$	C1
	$v = (V / R) / [(V / L) (N / V) e]$ or $(V / R) / [A (N / AL) e]$ = VL / RNe	A1
	or	
	$v = L / t$ = $L / (Q / I)$	(C1)
	= LI / Q = $L(V / R) / Ne$ = VL / RNe	(A1)

6(b)(iii)	time = distance / speed or Q / I $= L / (VL / RNe)$ or $Ne / (V / R)$ time = RNe / V	A1
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36.

5(a)	correct symbol for the heater or for the LDR	M1
	all correct symbols in series (ignore voltmeter) and no extra symbols	A1
	correct symbol for voltmeter and in parallel with the heater	B1
5(b)(i)	$R = \rho L / A$	C1
	$= (1.1 \times 10^{-6} \times 2.0) / 3.8 \times 10^{-7}$	A1
	$= 5.8 \Omega$	
5(b)(ii)	$I = 4.8 / 5.8$	A1
	$= 0.83 \text{ A}$	
5(c)(i)	A larger (for new wire) or $A \propto d^2$ (and d larger for new wire) or $R \propto 1 / d^2$ (and d larger for new wire)	M1
	so R is less (than that of first wire)	A1
5(c)(ii)	(heater / total resistance decreases so) current (in circuit) increases (so p.d. across LDR increases) or heater resistance decreases so it has a smaller share/proportion/fraction of the (total) voltage / e.m.f.	M1
	(so voltmeter) reading is less (than 4.8 V)	A1

37.

6(a)(i)	$(I =) 16 / 320 = 0.050 \text{ (A)}$	A1
6(a)(ii)	$R = (25 - 16) / 0.050$ or $R = (9 / 16) \times 320$ or $R = (25 / 0.050) - 320$	C1
	$R = 180 \text{ (}\Omega\text{)}$	C1
	$R_{\text{(LDR)}} = [(1 / 180) - (1 / 240)]^{-1}$ $= 720 \Omega$	A1
	or	
	$I = (25 - 16) / 240$ $(= 0.0375 \text{ A})$	(C1)
	$I_{\text{(LDR)}} = 0.050 - 0.0375$ $(= 0.0125 \text{ A})$	(C1)
	$R_{\text{(LDR)}} = 9.0 / 0.0125$ $= 720 \Omega$	(A1)

6(a)(iii)	$P = V^2 / R$ or $P = VI$ or $P = I^2 R$	C1
	ratio = $(9^2 / 720) / (9^2 / 240)$ or ratio = $(9 \times 0.0125) / (9 \times 0.0375)$ or ratio = $(0.0125^2 \times 720) / (0.0375^2 \times 240)$ ratio = $0.1125 / 0.3375$ = 0.33	A1
6(b)	resistance of LDR decreases	B1
	resistance of parallel combination decreases or total resistance (of circuit) decreases or current in resistor of resistance 320Ω increases or potential difference across parallel combination / LDR / 240Ω resistor decreases	M1
	voltmeter reading increases	A1

38.

6(a)	$R = \rho L / A$	C1
	$\rho = 0.92 \times 5.3 \times 10^{-7}$ = $4.9 \times 10^{-7} \Omega \text{ m}$	A1
6(b)(i)	(current =) $6.4 / 1600 = 4.0 \times 10^{-3} \text{ (A)}$	A1
6(b)(ii)	charge = $4.0 \times 10^{-3} \times 3.2 \times 60$ (= 0.768 C)	C1
	number = $0.768 / 1.6 \times 10^{-19}$ = 4.8×10^{18}	A1
6(b)(iii)	$6.4 / E = 1600 / (1400 + 1600)$ or $E = 6.4 + (4.0 \times 10^{-3} \times 1400)$ or $E = 4.0 \times 10^{-3} (1600 + 1400)$	C1
	$E = 12 \text{ V}$	A1
6(b)(iv)	$P = I^2 R$ or $P = VI$ or $P = V^2 / R$	C1
	ratio = $(6.4^2 / 1600) / [(4.0 \times 10^{-3})^2 \times 1400]$ = 1.1	A1
6(c)(i)	increase	B1
6(c)(ii)	increase	B1
6(c)(iii)	decrease	B1

39.

7(a)	current (through a conductor is directly) proportional to potential difference (across the conductor) or vice versa	M1
	(provided that) temperature (of conductor remains) constant	A1
7(b)(i)	$R = \rho L / A$	C1
	$\rho = (18 \times 7.2 \times 10^{-8}) / 0.94$ $= 1.4 \times 10^{-6} \Omega \text{ m}$	A1
7(b)(ii)	voltmeter reading = 3.1 V	A1
7(b)(iii)	current in the battery: increase	B1
	voltmeter reading: decrease	B1
7(b)(iv)	cross marked on the resistance wire to right of the arrowhead of S, but not touching the right-hand end of the resistance wire	B1
7(c)(i)	$I = Anvq$ $q = 0.93 / [(7.2 \times 10^{-8}) \times (9.0 \times 10^{28}) \times (1.3 \times 10^{-3})]$	C1
	$q = 1.1 \times 10^{-19} \text{ C}$	A1
7(c)(ii)	charge / q (value) is below $1.6 \times 10^{-19} \text{ (C)}$ or charge cannot be below $1.6 \times 10^{-19} \text{ (C)}$ or (the charge carriers / q) should have a charge of $1.6 \times 10^{-19} \text{ (C)}$	B1